MID-TERM/Minor EXAMINATION
(March 2022)

Subject Code: BAI-204	Subject: Optimization Techniques and Decision Making
Time: 1 ½ Hour	**Maximum Marks:** 30

Note: Q. 1 is compulsory. Attempt any two questions from the rest.

Q1 **(2.5*4)**

(a) What do you mean by optimization of a mathematical/logical problem? Suggest your thoughts.

(b) Define constraints in the optimization process with an example.

(c) Define at least four applications of optimization in the real world.

(d) Define feasible regions with an example.

Q2 **(5,5)**

(a) Consider a set of n points {(x1, y1), . . . ,(xn, yn)} in the plane. We want to find a point (x, y) such that the sum of the Euclidean distances from this point to all the other points are minimized.

Are the optimality conditions necessary? Sufficient? State clearly your assumptions.
Provide a semidefinite programming formulation of this problem

(b) Consider a set of n points {(x1, y1), . . . ,(xn, yn)} in the plane. We want to find a point (x, y) such that the sum of the Euclidean distances from this point to all the other points are minimized.

Is the objective function differentiable? Is this a convex optimization problem?
Write the corresponding optimality conditions.

Q3 **(5,5)**

(a) A computer maker ("XYZ, Inc.") produces 4000 desktops, 3000 laptops, and 9000 MP3 players per month. All models can be sold in either "standard" or "customized" configurations. The total number of items that can be customized during a normal month is equal to 2500. In addition, up to 1500 extra items can be customized on overtime, at a higher cost (increasing this would require negotiating with the workers' union). The net profits are as follows:

	Standard	Customized on Regular time	Customized on Overtime
Desktops	100	300	200
Laptops	120	400	230
MP3 Players	20	40	30

The objective is to find a production schedule that maximizes the total net profit.

Formulate this as a linear programming problem (not necessarily in standard form).

(b) Consider a set of n points {(x1, y1), . . . ,(xn, yn)} in the plane. We want to find a point (x, y) such that the sum of the Euclidean distances from this point to all the other points is minimized.

Give a nonlinear optimization formulation of this problem.

Q4 **(5,5)**

(a) You are organizing a workshop at IGTUW. The total number of students attending will be either 100 or 200 (with equal probability), but you will not find out the exact number until the day of the workshop. However, you need to commit today to firm dinner arrangements for all attendants. The cost of a restaurant dinner is the following: $30 per person for the first 150 guests, and $35 per person for every additional guest (over the first 150). Additionally, in case you cannot afford a restaurant dinner for everybody, you can order pizzas for the remaining guests at a cost of $5 per person. The total initial

budget is $3000.

Formulate an optimization problem, where you maximize the expected value of the number of guests that will enjoy a restaurant dinner at the workshop.

(b) You are organizing a workshop at IGTUW. The total number of students attending will be either 100 or 200 (with equal probability), but you will not find out the exact number until the day of the workshop. However, you need to commit today to firm dinner arrangements for all attendants. The cost of a restaurant dinner is the following: $30 per person for the first 150 guests, and $35 per person for every additional guest (over the first 150). Additionally, in case you cannot afford a restaurant dinner for everybody, you can order pizzas for the remaining guests at a cost of $5 per person. The total initial budget is $3000.

In this situation, what would happen if all 200 people show up? What happens if only 100 people show up? (the qualitative answer is OK).